IEEE HPEC 2026 · GRAPH ANALYTICS AND APPLICATION BENCHMARKING

Benchmarking GNN Inference on the Intel Core Ultra NPU: A Latency, Quantization, and Energy Analysis

Measurements on a power-constrained Meteor Lake client platform

The NPU accelerates regular FP32 vision models, but the iGPU is generally the better accelerator for the sparse GNN workloads evaluated here.

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Paper · code · data · figures

Term Notes

Why sparse GNNs challenge dense accelerators

What the NPU favors

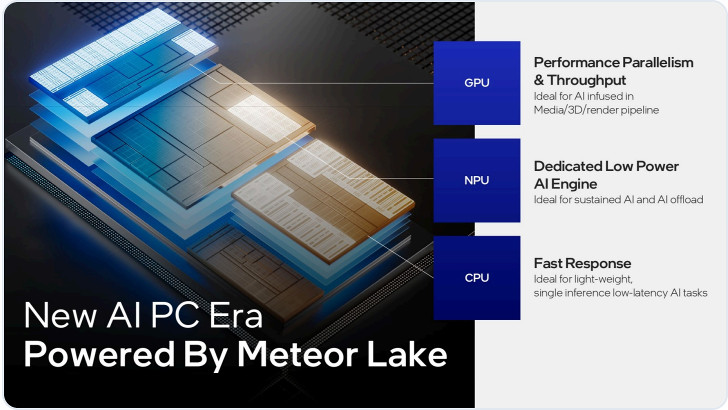
- Regular tensor shapes
- Predictable data movement
- Dense Conv and MatMul kernels
- Long fusion-friendly operator chains

What GNN inference introduces

- Sparse-times-dense multiplication (SpDMM)
- Gather, Scatter, and indirect indexing
- Low arithmetic intensity
- Operator and device-assignment uncertainty

Research question: when does a client NPU provide real end-to-end benefit rather than nominal device availability?

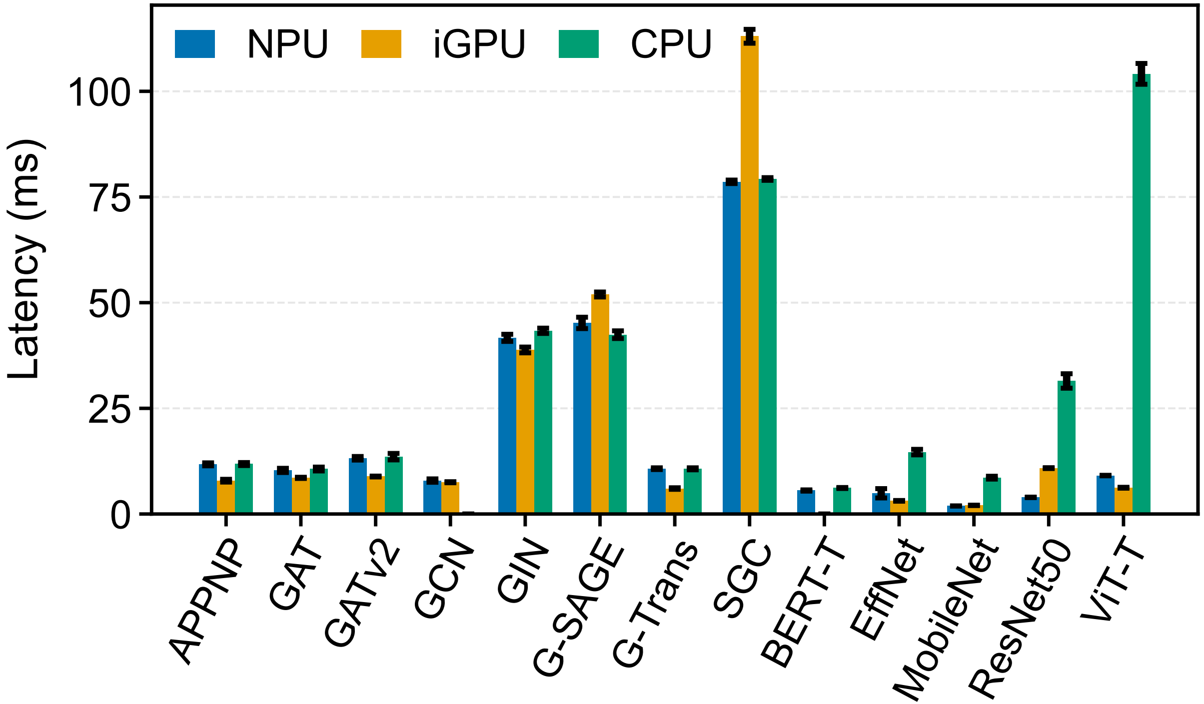
Measurement setup and protocol



Platform	Core Ultra 5 125H, 16 GB LPDDR5x, Windows 11
Backends	14-core CPU · Arc iGPU · AI Boost NPU 3720
Workloads	9 GNNs + 5 dense CNN/transformer baselines
Datasets	ogbn-arxiv · ogbn-products · ogbn-proteins
Protocol	Batch 1 · 5 warm-ups · 100 iterations · 3 runs
Paper stack	OpenVINO 2024.1 · ONNX Runtime 1.18

Timed latency includes provider dispatch and required host-device transfers after warm-up. Loading, preprocessing, conversion, and one-time compilation are excluded.

Dense models benefit; the iGPU leads the evaluated GNNs



Mean FP32 latency across the three OGB inputs; lower is better.

MobileNetV2:

1.90 ms on NPU

4.5x vs. CPU

ViT-Tiny:

9.10 ms on NPU

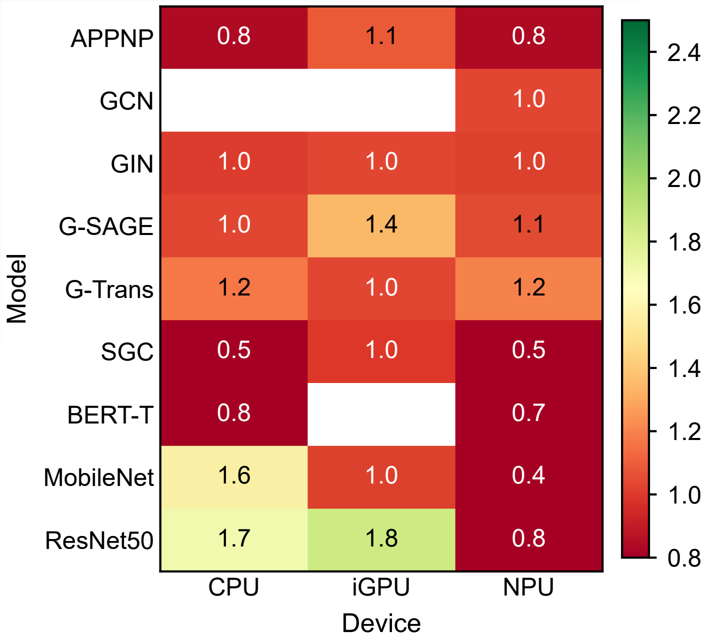
11.4x vs. CPU

GraphTransformer:

6.03 ms iGPU

10.72 ms NPU

INT8 is a compatibility decision, not a guaranteed speedup



Speedup below 1.0 means INT8 is slower than FP32.

Model	Requested	Outcome
SGC	NPU INT8	**2.2x slower**
MPNN	NPU FP32	CPU execution
BERT-Tiny	NPU INT8	Partial CPU execution
GAT / GATv2	NPU INT8	Compilation failed
EfficientNet-B0	NPU INT8	Compilation failed

Always inspect the execution trace: a requested NPU configuration does not prove native NPU execution.

Deployment guidance—and the boundary of the evidence

Use the measurements this way

- **Dense FP32 vision:** start with the NPU
- **Evaluated GNNs:** start with the iGPU
- **INT8:** benchmark per model and verify assignment
- **Energy:** do not infer isolated NPU power from package telemetry

Do not overgeneralize

- One Core Ultra 5 125H system
- One frozen paper software stack
- Established GNN reference workloads
- No post-quantization accuracy study

Explore the paper, code, CSV results, and reproduction workflow in the repository.

https://github.com/yusufarbc/intel-npu-gnn-benchmarking ↗